

HOMEWORK 7 FOR MATH 2860 (ELEMENTARY DIFFERENTIAL EQUATIONS)

E. A. ATINDAMA, PHD MATHEMATICS

Due at 10:00 EST in class

Question Q1. The complex-valued function $f(t)$ depends on real variable t :

$$f(t) = \exp [(-1 + i2\pi)t].$$

- (a) Compute $\operatorname{ref}(t)$ and $\operatorname{im}f(t)$.
- (b) Sketch $\operatorname{ref}(t)$ and $\operatorname{im}f(t)$ for $0 \leq t < 3$ (describe key features).
- (c) For which t is $f(t)$ purely real / purely imaginary?

Solution

(a) Write $f(t) = e^{-t}e^{i2\pi t} = e^{-t}(\cos(2\pi t) + i\sin(2\pi t))$. Hence

$$\operatorname{ref}(t) = e^{-t}\cos(2\pi t), \quad \operatorname{im}f(t) = e^{-t}\sin(2\pi t).$$

(b) Key features:

- Both are sinusoidal with period $T = 1$. They decay from value 1 at $t = 0$ because of the product with e^{-t} .
- Real part zeros:

$$\cos(2\pi t) = 0 \Rightarrow t = \frac{1}{4} + \frac{n}{2}, \quad n \in \mathbb{Z}.$$

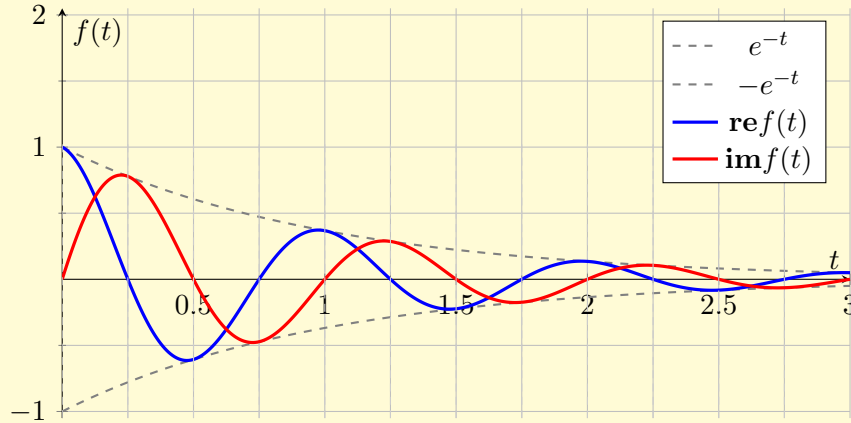
$$\text{In } [0, 3): t = \frac{1}{4}, \frac{3}{4}, \frac{5}{4}, \frac{7}{4}, \frac{9}{4}, \frac{11}{4}.$$

- Imaginary part zeros:

$$\sin(2\pi t) = 0 \Rightarrow t = \frac{n}{2}, \quad n \in \mathbb{Z}.$$

$$\text{In } [0, 3): t = 0, \frac{1}{2}, 1, \frac{3}{2}, 2, \frac{5}{2}.$$

- At $t = 0$: $\operatorname{ref}(0) = 1$, $\operatorname{im}f(0) = 0$. Both decay in amplitude as e^{-t} .



(c) Purely real / purely imaginary:

$$\operatorname{im}f(t) = 0 \iff \sin(2\pi t) = 0 \iff t = \frac{n}{2}, \quad n \in \mathbb{Z} \quad (\text{purely real}).$$

$$\operatorname{ref}(t) = 0 \iff \cos(2\pi t) = 0 \iff t = \frac{1}{4} + \frac{n}{2}, \quad n \in \mathbb{Z} \quad (\text{purely imaginary}).$$

Question Q2. Unforced oscillator:

$$m\ddot{x} + c\dot{x} + kx = 0, \quad m = 4, \quad k = 9, \quad c \geq 0.$$

- (i) Find characteristic roots.
- (ii) Identify ranges of c for (a) decaying without oscillation, (b) decaying with oscillation.
- (iii) Give one representative solution formula for each range.

Solution (minimal).

Characteristic equation:

$$4r^2 + cr + 9 = 0 \quad \Rightarrow \quad r = \frac{-c \pm \sqrt{c^2 - 144}}{8}.$$

Discriminant $\Delta = c^2 - 144$.

Ranges:

- $\Delta > 0$ (real distinct roots) $\iff c > 12$. $\Rightarrow$ decaying, no oscillation (overdamped).
- $\Delta = 0$ (double root) $\iff c = 12$. critically damped (decaying, no oscillation).
- $\Delta < 0$ (complex roots) $\iff 0 \leq c < 12$. $\Rightarrow$ decaying with oscillation (underdamped).

Representative solutions:

(a) Choose $c = 16 > 12$. Then

$$r = \frac{-16 \pm \sqrt{256 - 144}}{8} = \frac{-16 \pm \sqrt{112}}{8} = \frac{-16 \pm 4\sqrt{7}}{8} = \frac{-4 \pm \sqrt{7}}{2}.$$

General solution (overdamped):

$$x(t) = C_1 e^{\frac{-4+\sqrt{7}}{2}t} + C_2 e^{\frac{-4-\sqrt{7}}{2}t}.$$

(b) Choose $c = 8$ (underdamped, $0 \leq c < 12$). Then

$$r = \frac{-8 \pm \sqrt{64 - 144}}{8} = \frac{-8 \pm i\sqrt{80}}{8} = -1 \pm i\frac{\sqrt{5}}{2}.$$

General solution (decaying oscillation):

$$x(t) = e^{-t} \left(A \cos\left(\frac{\sqrt{5}}{2}t\right) + B \sin\left(\frac{\sqrt{5}}{2}t\right) \right).$$